TESTS FOR CATIONS

Positive ion in solution	***Effect of aqueous sodium hydroxide***	***Effect of aqueous ammonia***
Iron(III) (Fe^{3+})	Red-brown ppt., insoluble in excess	Red-brown ppt., insoluble in excess
Calcium (Ca^{2+})	White ppt., insoluble in excess	No ppt or very slight amount of ppt
Magnesium (Mg^{2+})	White ppt., insoluble in excess	White ppt., insoluble in excess
Zinc (Zn^{2+})	White ppt., soluble in excess, giving a colourless solution	White ppt., soluble in excess giving a colourless solution
Ammonium (NH_4^+)	Ammonia produced on warming	-
Copper(II) (Cu^{2+})	Light blue ppt., insoluble in excess	Light blue ppt., soluble in excess ammonia , giving a deep blue solution
Iron(II) (Fe^{2+})	Green ppt., insoluble in excess	Green ppt., insoluble in excess
Lead(II) (Pb^{2+})	White ppt., soluble in excess, giving a colourless solution	White ppt., insoluble in excess
Aluminium (Al^{3+})	White ppt., soluble in excess, giving a colourless solution	White ppt., insoluble in excess

Lead(II) ions can be distinguished from aluminium ions by insolubility of lead(II) chloride.

TESTS FOR ANIONS

Negative ion in solution	***Test***	***Test result***
Chloride (Cl^-)	Acidify with dilute nitric acid, then add aqueous silver nitrate	White ppt.
Iodide (I^-)	Acidify with dilute nitric acid, then add aqueous silver nitrate	Yellow ppt.
Sulfate (SO_4^{2-})	Acidify with dilute nitric acid, then add aqueous barium nitrate (or barium chloride)	White ppt.
Carbonate (CO_3^{2-})	Add dilute acid	Effervescence, carbon dioxide formed
Nitrate (NO_3^-)	Add aqueous sodium hydroxide then add aluminium foil; and warm carefully	Ammonia gas given off

TESTS FOR GASES

Gas	***Test and test result***
Ammonia (NH_3)	Turns damp red litmus paper blue
Carbon dioxide (CO_2)	Gives white ppt. with limewater (ppt. dissolves with excess CO_2)
Chlorine (Cl_2)	Bleaches damp litmus paper
Hydrogen (H_2)	“pop” with a lighted splint
Oxygen (O_2)	Relights a glowing splint
Sulfur dioxide (SO_2)	Turns aqueous acidified potassium manganate(VII) from purple to colourless